Metrics	Portfolio X	Portfolio Y	Benchmark
Portfolio return	8.35%	11.20%	9.55%
Standard deviation	7.50%	16.44%	11.98%
Sharpe ratio	0.71	0.50	0.55
Treynor ratio	0.05	0.06	0.05
Information ratio	(0.15)	0.31	—
Sortino ratio (MAR = 5%)	0.89	0.58	0.65

Question #1 - 4 of 54

Question ID: 1674949

Which portfolio had the *best* return to systematic risk ratio?

A) Benchmark.

B) Portfolio X.

C) Portfolio Y.

Question #2 - 4 of 54

Question ID: 1674950

Which statement is *most accurate* regarding Portfolios X and Y?

A) Portfolio X has an inferior reward/risk ratio as compared to Portfolio Y.

B) Portfolio Y outperformed the benchmark by taking greater risk.

C) Portfolio Y more closely resembles the benchmark as compared to Portfolio X.

Question #3 - 4 of 54

Question ID: 1674951

Which portfolio has the *best* return to downside risk ratio?

A) Portfolio Y.

B) Portfolio X.

C) Benchmark.

Question #4 - 4 of 54

Question ID: 1674952

An investor seeking to maximize her return and who is indifferent to short-term volatility should *most likely* invest in:

A) the benchmark.

B) Portfolio Y.

C) Portfolio X.

Question #5 of 54

Question ID: 1674927

For which of the following is a liability-based benchmark *most likely* appropriate?

A) A defined contribution employee retirement fund.

B) A defined benefit pension fund.

C) A philanthropic endowment with a 5% spending policy.

Question #6 of 54

Question ID: 1674902

In global performance evaluation, performance attribution seeks to:

A) measure the risk and return of the portfolio.

B) identify the sources of difference between portfolio and benchmark return.

C) differentiate whether returns come from a manager's luck or skill.

Question #7 of 54

Question ID: 1674907

A transactions-based attribution:

A) is more complicated than a holdings-based attribution.

B) uses the unadjusted portfolio assets from the beginning of a measurement period.

C) involves running regressions of broad market indices against portfolio returns to decompose investment performance.

Question #8 of 54

Question ID: 1674904

Which of the following is an example of macro attribution? Quantifying a:

A) portfolio manager's decision to bet against emerging-market currencies.

B) pension fund's decision to swap a private equity manager for a fixed-income manager because interest rates are low.

C) portfolio manager's decision to overweight telecom stocks because interest rates are low.

Question #9 of 54

Question ID: 1674908

Which of the following is *most correct* regarding an arithmetic attribution?

A) The arithmetic attribution is designed to explain the return differential between a portfolio and the S&P 500.

B) The arithmetic attribution necessitates adjustments to smooth the subperiod effects over longer periods.

C) Arithmetic attribution is better suited than geometric attribution for use with industry professionals.

Question #10 of 54

Question ID: 1674919

Sarah Cunningham, CFA, uses a top-down analysis process when considering new investments. In performing risk attribution analysis on an absolute basis, the *most appropriate* approach to measuring her performance should focus on:

A) attributing tracking error to relative allocation and selection.

B) each factor's marginal contribution to total risk and specific risk.

C) each security's marginal contribution to total risk.

Question #11 of 54

Question ID: 1674947

Which of the following measures would be the *most appropriate* one to use when comparing the results of two portfolios in which each portfolio contains many stocks from a broad selection of different industries?

A) Information ratio.

B) Sharpe ratio.

C) Treynor measure.

Question #12 of 54

Question ID: 1674943

The Treynor measure is correctly defined as a measure of a fund's:

A) return earned compared to its systematic risk.

B) excess earned compared to its systematic risk.

C) return earned compared to its unsystematic risk.

Question #13 of 54

Question ID: 1674901

An effective performance attribution process *most likely* includes:

A) the portfolio manager's prior decision-making process.

B) the active investment decisions taken by the portfolio manager.

C) a partial explanation of the portfolio's excess return and risk.

Question #14 of 54

Question ID: 1674899

What is the goal of performance appraisal?

A) Identification of overall risk and return.

B) Interpretation of performance attribution.

C) Identification of the sources of differences between portfolio and benchmark risk and return.

Question #15 of 54

Question ID: 1674906

Holdings-based attribution is *most appropriate* for assessing:

A) holdings data based on long time intervals.

B) passive index funds.

C) portfolios with high turnover.

Question #16 of 54

Question ID: 1674920

Lisa Roberts is trying to determine whether her equity investment managers have added any value, so she has decided to conduct a macro attribution analysis of their portfolio returns. Roberts has gathered the following results:

	Fund Weight	Fund Return	Benchmark Weight	Benchmark Return
Growth Manager	**40%**	9.75%	**35%**	8%
Small-cap growth	25%	12%	20%	8%
Mid-cap growth	15%	6%	15%	8%
Value Manager	**60%**	6%	**65%**	3.85%
Small-cap value	40%	7%	35%	2%
Mid-cap value	20%	4%	30%	6%
TOTAL	**100%**	**7.5%**	**100%**	**5.3%**

Given the results compiled, the allocation effect of Roberts's underweighting value in her portfolio using the Brinson-Fachler model is *closest* to:

A) −5 basis points.

B) 14 basis points.

C) 7 basis points.

Question #17 of 54

Question ID: 1674941

Which statement regarding the Sharpe ratio and Sortino ratio is *most accurate*?

A) In both the Sharpe ratio and Sortino ratio, a higher calculated ratio is better.

B) The Sortino ratio only considers upside risk, while the Sharpe ratio considers both upside and downside risk.

C) In both the Sharpe ratio and Sortino ratio, the risk-free rate is used as a return benchmark.

Question #18 of 54

Question ID: 1674932

Which of the following is the *most appropriate* method of calculating the manager's active return? The manager's active return is the:

A) market return minus the benchmark return.

B) portfolio return minus the benchmark return.

C) portfolio return minus the market return.

Question #19 of 54

Question ID: 1674914

For the purposes of risk attribution, an analysis of a top-down investor would *most likely* include:

A) tracking error resulting from overweighting the technology sector.

B) tracking error resulting from underweighting several stocks relative to the benchmark.

C) a security's marginal contribution to standard deviation.

Question #20 of 54

Question ID: 1674928

Which of the following is not regarded to be an essential characteristic of a valid benchmark?

A) Specified in advance.

B) Appropriate to the manager’s investment approach and style.

C) Reflective of past investment opinion.

Question #21 of 54

Question ID: 1674942

Analysis of a portfolio's drawdown is useful for evaluating:

A) how long it will take to recover after a period of losses.

B) a portfolio's downside capture relative to an index.

C) how much risk a portfolio manager is taking relative to returns.

Question #22 of 54

Question ID: 1674926

A liability-based benchmark is *most likely* to include:

A) nominal bonds only.

B) both nominal and inflation-adjusted bonds.

C) inflation-adjusted bonds only.

Question #23 of 54

Question ID: 1674933

XYZ Asset Management operates a Pan-European corporate bond fund. Last year, the bond fund generated a 5.1% return. In the same year, the bond fund's benchmark generated a 6.0% return; however, the broad index of European bonds generated a 4.5% return. XYZ's bond fund return due to active management is *closest* to:

A) −0.9%.

B) 0.6%.

C) 1.5%.

Question #24 of 54

Question ID: 1674929

Fund Sponsors often use the median account in a particular universe of account returns as an appropriate benchmark. This form of benchmark has numerous drawbacks. Which of the following is not a drawback that would be associated with using the median account as a benchmark?

A) It is not measurable as its value cannot be determined on a reasonably frequent basis.

B) As the median manager is unknown, the measure is ambiguous.

C) It is virtually impossible to identify the median manager in advance.

Question #25 of 54

Question ID: 1674900

Which component of performance evaluation determines whether the performance was affected primarily by skill or luck?

A) Performance appraisal.

B) Performance measurement.

C) Performance attribution.

Question #26 of 54

Question ID: 1674931

A retired investor needs to earn at least $100,000 per year in investment income to fund living expenses. Which type of benchmark is *best* suited for this individual?

A) Returns-based benchmark.

B) Absolute benchmark.

C) Broad market index.

TOPIC: PERFORMANCE EVALUATION

THE TOTAL POINT VALUE OF THIS QUESTION SET IS 12 POINTS

Johannes Holopainen is a performance attribution analyst at Tampere Research (TR), located in Finland. TR provides independent performance attributions of active fund managers on behalf of large institutional investors. Holopainen is asked by his manager, Emppu Turunen, to run a factor model attribution of the Wacken Fund, a fund managed by Islander Asset Management (IAM). The fund description claims to follow a semi-active investment style around a Carhart four-factor model approach. Turunen and his assistant, Tarja Hietala, prepare the data needed for the attribution analysis shown in Exhibit 1.

Exhibit 1: Data for Attribution Analysis

Factors	Portfolio	Benchmark	Difference	Factor Return	Absolute	Proportion of Total
RMRF	0.97	1.00		6.45%		
SMB	–1.08	–1.00		–4.25%		
HML	0.35	0.00		5.50%		

WML	–0.01	0.02		8.75%		
				Factor tilts return		
				Security selection return		
				Active return		

Turunen starts to complete the four-factor attribution shown in Exhibit 1 but is called away into a project meeting, so Turunen asks Hietala to complete the analysis. Turunen left a note that the Wacken Fund achieved an alpha of 1.359% over the measurement period.

Hietala completes the attribution analysis of the Wacken Fund started by Turunen.

Question #27 - 30 of 54

Question ID: 1674922

Based upon the data in Exhibit 1, which of the following statements is *least accurate*?

A) The benchmark is a broad-based index with average market risk.

B) The manager's large-cap tilt contributed negatively to returns.

C) The benchmark has a large-cap bias.

Question #28 - 30 of 54

Question ID: 1674923

Which of the following is *closest* to the return from security selection?

A) –0.35%.

B) +0.35%.

C) –0.45%.

Question #29 - 30 of 54

Question ID: 1674924

Which of the following is the proportion of total active return contributed by the value factor?

A) 141.65%.

B) 106.41%.

C) 1.925%.

Question #30 - 30 of 54

Question ID: 1674925

Which of the following would have had a greater positive impact on overall return compared to the manager's actual return?

A) Increasing exposure to the market factor to 1.06.

B) Increasing the stock selection return by +0.55%.

C) Increasing exposure to momentum to 0.04.

Question #31 of 54

Question ID: 1674905

Regression analysis is *most likely* to be used to analyze portfolio returns when performing:

A) returns-based attribution.

B) transactions-based attribution.

C) holdings-based attribution.

Question #32 of 54

Question ID: 1674910

Jonathan Kennedy is performing attribution analysis of a three-sector domestic equity portfolio. The following table shows data from the portfolio along with benchmark data.

Sector/Industry	Weight in Portfolio	Weight in Benchmark	Return From Portfolio	Return From Benchmark
Biotech	30%	25%	5%	3%
Utilities	40%	35%	10%	1%
Media	30%	40%	-5%	-4%
Total	100%	100%	-4%	-0.5%

Using the data in the table, the total selection effect is *closest* to:

A) 3.75%.

B) 4.25%.

C) 3.25%.

Question #33 of 54

Question ID: 1674939

One reason it can be difficult to benchmark hedge funds is because hedge funds:

A) frequently invest in illiquid assets.

B) generally perform in line with their peer universe.

C) generally underperform the broad market indices.

TOPIC: PORTFOLIO PERFORMANCE EVALUATION

TOTAL POINT VALUE OF THIS QUESTION SET IS 12 POINTS

Walker Nero is an investment analyst with Zimmer Advisors Ltd. (Zimmer). Nero's primary responsibility is evaluating the performance of investment managers being considered by Zimmer's clients. Grunder Preparatory School (Grunder) has hired Zimmer to advise the investment committee of its endowment fund.

Grunder has most of its portfolio in large-cap equities, and the investment committee realizes the investment strategy is extremely competitive—so it asks Zimmer to evaluate its large-cap manager compared to the manager's competitors. As a result, Nero benchmarks the performance of Grunder's large-cap equity manager against the median manager from the manager's peer group, using data provided by a popular investment consultant.

Grunder's investment committee then asks Nero to evaluate the performance of the Knight Fund. It also asks for advice regarding additional metrics that can be used in evaluating manager performance. The most recent risk and return measures for the Knight Fund are shown in the following table. The T-bill return over the same time period was 3.0%. The return on the S&P 500 was used as the market index.

	Knight Fund	Market Index
Return	6.80%	6.60%
Beta	0.90	1.00
Standard deviation	32.60%	31.00%

Question #34 - 37 of 54

Question ID: 1674954

Which of the following statements regarding the use of the median manager's results as a benchmark for evaluating performance of a portfolio is *most accurate*?

A) It is not possible to identify the median manager.

B) The weights of individual securities required to replicate the median manager cannot be known.

C) The performance of the median manager is subject to upward bias.

Question #35 - 37 of 54

Question ID: 1674955

The Treynor ratio for the Knight Fund is *closest* to:

A) 7.56.

B) 0.90.

C) 4.22.

Question #36 - 37 of 54

Question ID: 1674956

The Sharpe ratio for the Knight Fund is *closest* to:

A) 0.90.

B) 0.21.

C) 0.12.

Question #37 - 37 of 54

Question ID: 1674957

Which performance measure would Nero choose if Nero wanted to consider the impact of both market-related variability of returns and the variability of active management?

A) Treynor ratio.

B) Information ratio.

C) Sharpe ratio.

Question #38 of 54

Question ID: 1674909

The *most common* approach for calculating excess returns for individual investor portfolios involves calculating:

A) arithmetic attribution results.

B) total return attribution results.

C) geometric attribution results.

Question #39 of 54

Question ID: 1674911

Ha-Yoon Kim is a stock analyst with Yeongli Securities Co. She uses the Carhart factor model to perform attribution analysis. Output from the model indicates the benchmark that Kim uses has the following factor sensitivities:

Factor	Benchmark
RMRF	+1.00
SMB	−1.00
HML	0.05
WML	−0.95

Assuming that she is using a benchmark that most closely resembles her portfolio attributes, we can conclude that Kim is *most likely* a:

A) large-cap investor.

B) small-cap investor.

C) momentum investor.

Question #40 of 54

Question ID: 1674946

Which of the following statements regarding the Sharpe ratio is *most accurate*?

A) Beta is not a component of the Sharpe ratio.

B) The measure of risk used in the denominator of the Sharpe ratio is standard deviation, also known as unsystematic risk.

C) The denominator of the Sharpe ratio is standard deviation, which comprises partly systematic risk called beta.

Use the following table for Questions 8 through 10.

	Fund Weight	Fund Return	Benchmark Weight	Benchmark Return
Domestic equities	0.50	5.78%	0.75	6.20%
International equities	0.50	8.14%	0.25	8.51%
Total	1.00	6.96%	1.00	6.78%

Question #41 - 43 of 54

Question ID: 1674916

Based on the Brinson-Fachler model, the allocation effect of hiring the domestic equities portfolio manager is *closest* to:

A) −0.15%.

B) 0.15%.

C) 0.11%.

Question #42 - 43 of 54

Question ID: 1674917

The selection and interaction effect of hiring the international equities portfolio manager is *closest* to:

A) 0.00%.

B) −0.19%.

C) 0.09%.

Question #43 - 43 of 54

Question ID: 1674918

Which statement *most accurately* describes the impact from asset allocation and manager selection in the previous table?

A) Value was added with manager selection, and value was added with asset allocation.

B) Value was detracted by manager selection, but value was added with asset allocation.

C) Value was added with manager selection, but value was detracted with asset allocation.

Question #44 of 54

Question ID: 1674945

Peter Michaels, CFA, works at Composite Consulting, and is in charge of evaluating the performance of various portfolio managers. His main tasks are to measure and evaluate the sources of return that can be attributed to manager performance. Michaels understands the importance of incorporating risk into his analyses, but he realizes the limitations associated with some performance measurement techniques in accomplishing that particular objective. Michaels begins the evaluation of numerous managers by examining return information from both the portfolio being evaluated and its designated benchmark.

Michaels has the following return information for the AM Growth Fund:

	AM Growth Fund	S&P 500
Return	14%	12%
Standard deviation	25%	18%
Beta	1.15	1.00

If the AM Growth Fund is considered to be well diversified, which measure would be *most appropriate* in evaluating its risk/return performance?

A) The Treynor measure.

B) The M^2 measure.

C) The Sharpe ratio.

Question #45 of 54

Question ID: 1674944

Which of the following statements about risk/return investment manager performance measures is *least accurate*?

A) The Treynor measure includes company-specific risk as part of its performance measurement.

B) The Sharpe measure includes company-specific risk as part of its performance measurement.

C) When measuring the performance of an equity fund, if the Sharpe ratio is 0.55, and the Treynor measure is 0.47, the difference is attributable to unsystematic (company-specific) risk.

Santiago Barrera is a performance evaluation analyst with Zapopan Asset Management (ZAM). ZAM provides performance attribution analysis services to many large institutional investors across the Americas.

Barrera has been asked by his manager, Mateo Garcia, to evaluate the performance of three U.S. funds: Delta, Gamma, and Theta. Delta has a U.S. small-cap style, Gamma a U.S. momentum style, and Theta has a U.S. large-cap growth style.

Garcia explains that the client is interested in the separation of true active returns from style returns in the analysis. Jose Reynosa, a junior analyst at ZAM, assists Barrera in compiling relevant performance data, as shown in Exhibit 1.

Exhibit 1: Performance Data

Returns	Delta	Gamma	Theta
Portfolio returns	9.2%	–5.8%	6.7%
Risk-free		3%	
S&P 500		8.8%	
S&P 500 Growth		9.5%	
S&P SmallCap 600		8.3%	
S&P 500 Momentum		–7.1%	

Question #46 - 49 of 54

Question ID: 1674935

Rank the true active returns from the three portfolios from *highest* to *lowest*.

A) Gamma, Theta, Delta.

B) Gamma, Delta, Theta.

C) Theta, Delta, Gamma.

Question #47 - 49 of 54

Question ID: 1674936

Calculate the style return for the Delta portfolio.

A) 0.90%.

B) 0.40%.

C) –0.50%.

Question #48 - 49 of 54

Question ID: 1674937

Which of the portfolios has misfit active return of 0.70%?

A) Delta.

B) Gamma.

C) Theta.

Question #49 - 49 of 54

Question ID: 1674938

Which manager's normal portfolio return is the *highest*?

A) Gamma.

B) Delta.

C) Theta.

Question #50 of 54

Question ID: 1674940

In the Sortino ratio, the excess return is divided by the:

A) standard deviation using only the returns below a minimum level.

B) maximum drawdown.

C) standard deviation.

Question #51 of 54

Question ID: 1674913

The following table shows attribution results from a fixed-income portfolio:

Duration Bucket	Sector	Duration Effect	Curve Effect	Total Interest Rate Allocation Effect	Sector Allocation	Bond Selection	Total
Short	Corporate						
	ABS/MBS						
	Total	−0.25%	+0.45%	+0.20%	+0.75%	+0.15%	+1.10%
Long	Corporate						
	ABS/MBS						
	Total	−0.40%	+0.05%	−0.35%	+0.15%	−0.45%	−0.65%
TOTAL		**−0.65%**	**+0.50%**	**−0.15%**	**+0.90%**	**−0.30%**	**+0.45%**

Based on the results provided in this table, it would be *most appropriate* to conclude that:

A) the portfolio outperformed the benchmark portfolio by 45 basis points.

B) the yield curve likely flattened, because long- and short-duration positions underperformed the benchmark.

C) there was a 90 basis point gain from underweighting ABS/MBS versus corporate bonds.

Question #52 of 54

Question ID: 1674903

Which of the following is *least likely* to represent a benefit of carrying out performance attribution?

A) Additional information for passive portfolios.

B) Quality control of active portfolios.

C) Justification of underperformance during negative market periods.

Question #53 of 54

Question ID: 1674912

Within yield curve decomposition analysis, which of the following statements regarding the full-repricing method is *least accurate*? The full-repricing method:

A) requires less data than the duration-based approach.

B) is most accurate when based on spot rates.

C) is most accurate when based on zero-coupon curves.

Question #54 of 54

Question ID: 1674930

Which of the following *best* characterizes manager universes as a benchmark? Manager universes are:

A) a valid benchmark because they are measurable.

B) not a valid benchmark because they are not investable.

C) not a valid benchmark because they are not measurable.